AGENTSHIELD

AI Agent Security & Runtime Governance | Solution Pitch

A security gateway between AI agents and sensitive tools - so an agent can propose an action, but policy decides whether it may happen.

The problem

AI agents are increasingly connected to tools that can change business data or trigger consequential actions. A model response alone should not be treated as authorization: a mistaken, manipulated, or poorly verified request could lead to an unauthorized refund, data exposure, or other harmful operation.

Our solution

AgentShield is a runtime security and governance platform that evaluates protected agent actions before execution. It separates the agent's reasoning from the authority to act. Structured action proposals are checked against schemas, identity and resource ownership, configured policies, and risk conditions. Depending on the result, AgentShield allows the action, blocks it, or requires additional verification or human review. Important decisions are recorded in a tamper-evident audit chain.

How it works - customer support demonstration

1. Request	A customer reports a product issue and asks for a refund.
2. Verify	The support workflow checks customer ownership, order status, payment and delivery state, issue type, and configured return rules.
3. Decide	The security gateway applies action policies and may deny, require step-up verification, or send the refund for human approval.
4. Record	Only an authorized refund reaches the simulated service; the decision and execution are captured in the audit log.

Security capabilities

Deterministic policy decisions (ALLOW / DENY / VERIFY) • schema and ownership checks • one-time, parameter-bound step-up challenges • prompt-injection pattern detection, including English and Hinglish examples • PII scrubbing and response egress filtering • voice/ASR trust gating for sensitive actions • request-velocity protections • hash-chained audit events • provider timeout and budget safeguards.

Implementation and scope

The demo uses a React dashboard, an Express API, and a local SQLite simulation, with a Gnani provider adapter and an offline-friendly demo mode. Customer-support classification and troubleshooting are deterministic demo logic rather than a live LLM support conversation. Banking, orders, and refunds are simulated: no real payment, commerce, or external business system is connected. The project is a prototype and is not production-ready authentication or deployment infrastructure.

Project repository https://github.com/OmAmbole009/agentshield

Core value AI can recommend; policy controls execution.